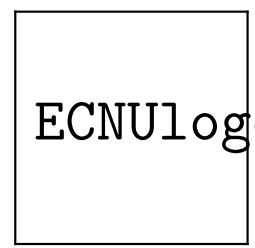
Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

PENDAHULUAN
TINJAUAN PUSTAKA
[progressbar=frametitle,
numbering=fraction]beamerthememetropolis
appendixnumberbeamer
booktabs [scale=2]ccicons
pgfplots dateplot
xspace
[fontset=ubuntu]ctex
[]beamerfontthemeprofessionalfonts mathspec [BoldFont=Fira
Sans, Numbers=OldStyle]Fira Sans Light
(Digits)[Numbers=Lining, Proportional]Fira Sans Light
xcolor
[BoldFont=Fira Sans SemiBold]Fira Sans Book


ECNUlogo.png

1. Introduction

2. Titleformats
   Tricks

3. Elements

4. Conclusion

Metropolis

The **metropolis**theme is a Beamer theme with minimal visual noise inspired by the hsrm Beamer Theme by Benjamin Weiss.
Enable the theme by loading tm

```
\documentclass{beamer}
\usetheme{metropolis}
```

Note, that you have to have Mozilla's *Fira Sans* font and XeTeX installed to enjoy this wonderful typography.

Sections

Sections group slides of the same topic

```
\section{Elements}
```

for which **metropolis**provides a nice progress indicator . . .

Metropolis titleformats

metropolissupports 4 different titleformats:

- Regular
- Smallcaps
- allsmallcaps
- ALLCAPS

They can either be set at once for every title type or individually.

titleformat frame=smallcaps

Small caps

This frame uses the `smallcaps` titleformat.

Potential Problems

Be aware, that not every font supports small caps. If for example you typeset your presentation with pdfTeX and the Computer Modern Sans Serif font, every text in smallcaps will be typeset with the Computer Modern Serif font instead.

titleformat frame=allsmallcaps

All small caps

This frame uses the `allsmallcaps` titleformat.

Potential problems

As this titleformat also uses smallcaps you face the same problems as with the `smallcaps` titleformat. Additionally this format can cause some other problems. Please refer to the documentation if you consider using it.
As a rule of thumb: Just use it for plaintext-only titles.

titleformat frame=allcaps

All caps

This frame uses the `allcaps` titleformat.

Potential Problems

This titleformat is not as problematic as the `allsmallcaps` format, but basically suffers from the same deficiencies. So please have a look at the documentation if you want to use it.

Typography

```
The theme provides sensible defaults to
\emph{emphasize} text, \alert{accent} parts
or show \textbf{bold} results.
```

becomes

The theme provides sensible defaults to *emphasize* text, accent parts or show **bold** results.

Font feature test

- Regular
- *Italic*
- SmallCaps
- **Bold**
- ***Bold Italic***
- **Bold SmallCaps**
- `Monospace`
- *`Monospace Italic`*
- `Monospace Bold`
- *`Monospace Bold Italic`*

Lists

Items

- Milk
- Eggs
- Potatos

Enumerations

1. First,
2. Second and
3. Last.

Descriptions

PowerPoint Meeh.

Beamer Yeeeha.

Animation

- This is important

Animation

- This is important
- Now this

Animation

- This is important
- Now this
- And now this

Animation

- This is really important
- Now this
- And now this

Figures

coordinate](0,0)coordinate(A)++(90:
5cm)coordinate(B)++(0:5cm)coordinate(C)++(-90:
5cm)coordinate(D);\drawfill=alerted text.fg!20] (A) – (B) – (C)
–(D) – cycle; in 1,...,40
coordinate]coordinate(X)at(A){};\pathcoordinate] (A) – (B)
coordinate[pos=.10](A) – (C) coordinate[pos=.10](B) – (D)
coordinate[pos=.10](C) – (X) coordinate[pos=.10](D); [fill=alerted
text.fg!40] (A)–(B)–(C)– (D) – cycle;

Gambar: Rotated square from texample.net.

Tables

Tabel: Largest cities in the world (source: Wikipedia)

City	Population
Mexico City	20,116,842
Shanghai	19,210,000
Peking	15,796,450
Istanbul	14,160,467

Blocks

Three different block environments are pre-defined and may be styled with an optional background color.

Default

Block content.

Alert

Block content.

Example

Block content.

block=fill

Default

Block content.

Alert

Block content.

Example

Block content.

Math

block=fill

Theorem
$\int_a^b f(x)\mathrm{d}x = F(b) - F(a)$

Bukti.
$g(x) = e^x - x - 1 \; x > 1 \quad g'(x) = e^x - 1 > 0 \quad g(x) > g(1) = 0$
$x > 1 \quad e^x > 1 + x$ □

$$e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n$$

Line plots

[mlineplot, width=0.9height=6cm,]
sin(deg(x)); +[samples=100] sin(deg(2*x));

Bar charts

[mbarplot, xlabel=Foo, ylabel=Bar, width=0.9height=6cm,]
plot coordinates (1, 20) (2, 25) (3, 22.4) (4, 12.4); plot coordinates (1, 18) (2, 24) (3, 23.5) (4, 13.2); plot coordinates (1, 10) (2, 19) (3, 25) (4, 15.2);
lorem, ipsum, dolor

Quotes

Veni, Vidi, Vici

Frame footer

metropolisdefines a custom beamer template to add a text to the footer. It can be set via

```
\setbeamertemplate{frame footer}{My custom footer}
```

References

Some references to showcase [allowframebreaks] [?, ?, ?, ?, ?]

Summary

Get the source of this theme and the demo presentation from

github.com/matze/mtheme

The themeitself is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License.

Backup slides

Sometimes, it is useful to add slides at the end of your presentation to refer to during audience questions.
The best way to do this is to include the `appendixnumberbeamer` package in your preamble and call `\appendix`

References I